METRIC

DOD-D-51524(EA)
20 May 1986

MILITARY SPECIFICATION

DYE, DISPERSE BLUE 3 (METRIC)

This specification is approved for use within the Chemical Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Disperse Blue 3 dye, chemically known as 1-methylamino-4-ethanolaminoanthraquinone.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

L-P-378 - Plastic Sheet and Strip, Thin Gauge, Polyolefin
PPP-D-723 - Drums, Fiber

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-SPD-TS, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. :

AMSC N/A

FSC 6820

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:03Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-D-3464 - Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification

STANDARDS

FEDERAL

Fed. Std. No. 313 - Material Safety Data Sheets Preparation and the Submission of

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS shall be the issue of nongovernment documents which is current on the date of the solicitation.

ASTM STANDARDS

D 1895 - Apparent Density, Bulk Factor, and Pourability of Plastic Materials (DOD Adopted)
D 3951 - Commercial Packaging (DOD Adopted)
E 11 - Wire-Cloth Sieves for Testing Purposes (DOD Adopted)

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specification, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:03Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Chemical and physical characteristics. Dye, Disperse Blue 3 shall conform to the chemical and physical characteristics of table I when tested as specified therein.

TABLE I. Chemical and physical characteristics

Characteristic	Requirement	Test paragraph
Purity, percent by weight, minimum	96.0	4.2.4.1
Particle size, percent by weight passing:		4.2.4.2
No. 20 sieve, minimum	99.9	
No. 50 sieve, minimum	95.0	
No. 325 sieve, maximum	40.0	
Apparent density, grams per milliliter, minimum	0.20	4.2.4.3

3.2 Material Safety Data Sheets. Material Safety Data Sheets for Dye, Disperse Blue 3 shall be prepared and submitted by the contractor as specified in Fed. Std. No. 313 (see 6.3).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:03Z
Check the source to verify that this is the current version before use.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the Dye, Disperse Blue 3 offered for acceptance at one time which has been produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation (see 6.4), each batch shall constitute a lot.

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For test. Three representative specimens of approximately 200 grams (g) each shall be randomly removed from the lot or batch of dye offered for acceptance and shall be placed in separate clean, dry containers labeled to identify the lot represented.

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled and closed shipping container, ready for shipment. Sample containers shall be examined for the following defects using an AQL of 2.5 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Polyethylene liner or closure thereof not as specified (level B only)
(d) Container closure not as specified
(e) Container damaged or leaking
(f) Marking incorrect, missing, or illegible
(g) Desiccant not as specified or missing (level B only)
(h) Unitization not as specified

4.2.3.2 For test. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.4 Tests. Tests shall be conducted as follows:

4.2.4.1 Purity. Use only high purity grades of methanol and water suitable for liquid chromatography residue analysis. Prepare and measure the area of the chromatographic peaks of a standard Disperse Blue 3 dye solution (see 6.5) and a solution of the specimen as follows: Weigh to the nearest 0.1 milligram (mg) exactly 100.0 mg of the dye and transfer quantitatively to a 200-milliliter (mL) volumetric flask. Dissolve the dye in a solvent consisting of 90 percent by volume of methanol and 10 percent by volume of water, mix thoroughly, and dilute to 200 mL with the solvent (this stock solution contains 0.5

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:03Z
Check the source to verify that this is the current version before use.

mg of dye per milliliter). Using a suitable high pressure liquid chromatograph, inject 10 microliters of the solutions and using a suitable integrator, measure the area under the Disperse Blue 3 peaks with the detector set at 292 nanometers. (Disperse Blue 3 dye is a mixture of three isomeric compounds. Retention times are approximately 10.1, 8.87, and 8.03 minutes; actual retention times will depend on the age of the column. A better measure will be the relative retention times. Setting the third component relative retention time (RRT) to 1.0, the first component RRT should be calculated to be 0.803, and the second component RRT should be calculated to be 0.880.) The mobile phase is methanol/water (90/10) flowing at 0.4 mL per minute (20 megapascals) through a 2.5 by 70 millimeter (mm) guard column and a 4.6 by 250 mm analytical column, both containing "Partisil-10 ODS" (used for reversed-phase chromatography) (see 6.6). Calculate the percent by weight concentration of each specimen isomeric peak as follows:

$$P_n = \text{Percent purity of component } n = \frac{A_n C_n}{B_n}$$

where: $n = 1$ to 3 and

A_n = Area of specimen peak for component n,
B_n = Area of standard peak for component n, and
C_n = Percent by weight purity of standard component n.

Report the percent purity as the average of four sets of data from repeat injections. Sum the data for peaks 1, 2, and 3 for each injection to get percent dye purity for that injection.

4.2.4.2 Particle size. Use sieves conforming to ASTM E 11. Nest No. 20, No. 50, and No. 325 sieves in order of decreasing mesh size (with the sieve of largest mesh on top) on a receiving pan. Weigh to the nearest 0.1 g approximately 10 g of the specimen onto the top sieve and brush the material with a camel's-hair brush until no more specimen passes through the sieve. Remove the sieve and weigh the material retained. Repeat this procedure with each of the other sieves. Calculate the percent by weight material passing through each sieve.

4.2.4.3 Apparent density. Determine the apparent density in accordance with ASTM D 1895, method A.

5. PACKAGING

5.1 Unit packing. Dye, Disperse Blue 3 shall be unit packed level B or commercial as specified (see 6.2).

5.1.1 Level B. Uniform quantities of no less than 65 and no more than 130 kilograms ± 1/4 percent of the marked net weight of dye shall be packed in a fiber drum conforming to type III, grade A, class 2 of PPP-D-723. Each drum shall be provided with a close-fitting bag liner formed from polyethylene of

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:03Z
Check the source to verify that this is the current version before use.

101.6 micrometers minimum thickness conforming to type I, class 1, grade and finish optional of L-P-378. Seams shall be completely heat sealed and shall meet the heat-seal strength requirements of L-P-378. The filled bag shall be closed by heat sealing, tying, or knotting. A minimum of eight units of desiccant conforming to MIL-D-3464 shall be placed on top of the closed liner. Drum closure shall be fully gasketed and shall form a tight seal between the rim and the top of the drum.

5.1.2 Commercial. Uniform quantities of no less than 65 and no more than 130 kilograms ± 1/4 percent of the marked net weight of dye shall be unit packed in accordance with ASTM D 3951.

5.2 Packing. Dye, unit packed as specified in 5.1, shall require no further protection for shipment other than unitization.

5.3 Unitization. Shipping containers shall be unitized in accordance with the applicable requirements of MIL-STD-147 except that the pallet shall be as specified in the contract (see 6.2).

5.4 Marking. Marking for level B shipments shall be in accordance with MIL-STD-129. Marking for commercial shipments shall be in accordance with ASTM D 3951. All shipments shall be marked to show lot number and date of manufacture. Each container shall be durably and legibly marked with contrasting letters and background to show the net weight of contents. In addition, each container of material shall be marked in accordance with the requirements of public law and regulations. The marking shall include as applicable: name of product; quantity; warning symbol; signal word designating degree of hazard; actions to be followed or avoided; instructions in case of contact or exposure and notes to physicians; instructions in case of fire, spill, or leak; instructions for handling and storage; and disposal instructions.

6. NOTES

6.1 Intended use. Dye, Disperse Blue 3 is intended for use in the manufacture of colored signaling smoke mixtures.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification
(b) Level of unit packing required (see 5.1)
(c) Type of pallet required (see 5.3)

6.3 Material Safety Data Sheets. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with Fed. Std. No. 313. The pertinent mailing addresses for submission of the data are listed in appendix B of Fed. Std. No. 313.

6.4 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:03Z
Check the source to verify that this is the current version before use.

6.5 Dye of known purity. Dye, Disperse Blue 3 of known purity for use in 4.2.4.1 may be obtained by authority of the Procuring Contracting Officer (PCO). Alternately, the contractor may prepare his own standard dye and obtain verification of its purity by authority of the PCO from Commander, U.S. Army Armament, Munitions and Chemical Command, Product Assurance Directorate, Aberdeen Proving Ground, MD 21010-5423.

6.6 Trademark. "Partisil" is a trademark of Whatman Inc. "Partisil-10 ODS" is available from Whatman Inc., Chemical Separation Division, 9 Bridewell Place, Clifton, NJ 07014.

6.7 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E 29.

6.8 Colour Index. The Colour Index number of Dye, Disperse Blue 3 is C.I. 61505.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 6820-A092

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:03Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 4966 Alexandria, VA

POSTAGE WILL BE PAID BY

Commander
U.S. Army Chemical Research, Development and
Engineering Center
ATTN: SMCCR-SPD-TS
Aberdeen Proving Ground, MD 21010-5423

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:03Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
DOD-D-51524(EA)	DYE, DISPERSE BLUE 3 (METRIC)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:03Z
Check the source to verify that this is the current version before use.